

Juno (spacecraft)

Juno



Artist's rendering of the *Juno* spacecraft

Names	New Frontiers 2
Mission type	Jupiter orbiter
Operator	NASA / Jet Propulsion Laboratory
COSPAR ID	2011-040A (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2011-040A)
SATCAT no.	37773
Website	www.nasa.gov/mission_pages/juno/main/index.html (https://www.nasa.gov/mission_pages/juno/main/index.html) (NASA) www.missionjuno.swri.edu (https://www.missionjuno.swri.edu/) (SwRI)
Mission duration	Total: 7 years (planned) 14 years, 300 days (elapsed) Cruise: 4 years, 335 days Prime mission: 2 years (planned) 5 years, 16 days Extended mission: 4 years, 315 days (elapsed)^[a]

Spacecraft properties

Manufacturer Lockheed Martin Space

Launch mass	3,625 kg (7,992 lb) ^[1]
Dry mass	1,593 kg (3,512 lb) ^[2]
Dimensions	20.1 × 4.6 m (66 × 15 ft) ^[2]
Power	14 kW at Earth, ^[2] 435 W at Jupiter ^[1] 2 × 60-ampere hour, 28 Volt lithium-ion batteries^[3]

Start of mission

Launch date	August 5, 2011, 16:25:00 UTC
Rocket	Atlas V 551 (AV-029)
Launch site	Cape Canaveral, SLC-41
Contractor	United Launch Alliance

Flyby of Earth

Closest approach	October 9, 2013
Distance	559 km (347 mi)

Jupiter orbiter

Orbital insertion	July 5, 2016, ^[4] 9 years, 10 months, 26 days ago
Orbits	76 (planned) ^{[5][6]}

Orbital parameters

Perijove altitude	4,200 km (2,600 mi) altitude
Apojove altitude	75,600 km (47,000 mi) radius
Inclination	8.1 × 10 ⁶ km (5.0 × 10 ⁶ mi)
Inclination	90° (polar orbit)

Instruments



Juno mission patch

New Frontiers

Juno is a NASA space probe orbiting the planet Jupiter. Built by Lockheed Martin and operated by NASA's Jet Propulsion Laboratory, the spacecraft was launched from Cape Canaveral Air Force Station on August 5, 2011 UTC, as part of the New Frontiers program.^[7] *Juno* entered a polar orbit of Jupiter on July 5, 2016, UTC,^{[5][8]} to begin a scientific investigation of the planet.^[9] After completing its mission, *Juno* was originally planned to be intentionally deorbited into Jupiter's atmosphere,^[9] but has since been approved to continue orbiting until

contact is lost with the spacecraft.^[10] It will continue to explore Jupiter to study Jovian rings and inner moons area which is not well explored;^[11] this phase will also include close flybys of the moons Thebe^[12], Amalthea, Adrastea, and Metis.^[13]

Juno's mission is to measure Jupiter's composition, gravitational field, magnetic field, and polar magnetosphere. It also searches for clues about how the planet formed, including whether it has a rocky core, the amount of water present within the deep atmosphere, mass distribution, and its deep winds, which can reach speeds up to 620 km/h (390 mph).^[14]

Juno is the second spacecraft to orbit Jupiter, after the RTG-powered Galileo orbiter, which orbited from 1995 to 2003. Unlike all earlier spacecraft sent to the outer Solar System and beyond—which used radioisotope thermoelectric generators for power—*Juno* is powered by solar panels, more commonly used by satellites orbiting Earth and working in the inner Solar System.^[9] Accordingly, *Juno* required the three largest solar panel wings ever deployed on a planetary probe (at the time of launching). These play an integral role in stabilizing the spacecraft as well as generating power.^[15]

As of February 2026, Juno remained operational and in contact with Earth through the NASA Deep Space Network.^[16]

Naming

Juno's name comes from Greek and Roman mythology. The god Jupiter drew a veil of clouds around himself to hide his mischief, and his wife, the goddess Juno, was able to peer through the clouds and reveal Jupiter's true nature.^[17]

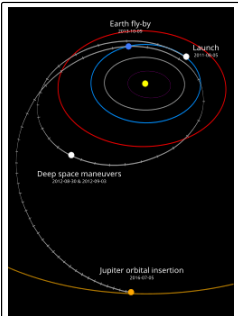
A NASA compilation of mission names and acronyms referred to the mission by the backronym Jupiter Near-polar Orbiter.^[18] However the project itself has consistently described it as a name with mythological associations^[19] and not an acronym. The spacecraft's current name is in reference to the Roman goddess Juno.^[17] *Juno* is sometimes called the *New Frontiers 2* as the second mission in the New Frontiers program,^{[20][21]} but is not to be confused with New Horizons 2, a proposed but unselected New Frontiers mission.

Overview

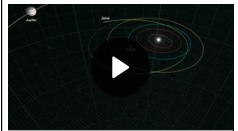
Juno was selected on June 1, 2005, as the next New Frontiers mission after *New Horizons*.^[22] The desire for a Jupiter probe was strong in the years prior to this, but there had not been any approved missions.^{[23][24]} The Discovery Program had passed over the somewhat similar but more limited Interior Structure and Internal Dynamical Evolution of Jupiter (INSIDE Jupiter) proposal,^[24] and the turn-of-the-century era Europa Orbiter was canceled in 2002.^[23] The flagship-level Europa Jupiter System Mission was in the works in the early 2000s, but funding issues resulted in it evolving into ESA's Jupiter Icy Moons Explorer.^[25]



Juno in launch configuration



Juno's interplanetary trajectory; tick marks at 30-day intervals.



Juno spacecraft trajectory animation

Juno completed a five-year cruise to Jupiter, arriving on July 5, 2016.^[8] The spacecraft traveled a total distance of roughly 2.8×10^9 km (19 AU; 1.7×10^9 mi) to reach Jupiter.^[26] The spacecraft was designed to orbit Jupiter 37 times over the course of its mission. This was originally planned to take 20 months.^{[5][6]}

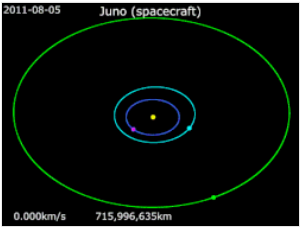
Juno's trajectory used a gravity assist speed boost from Earth, accomplished by an Earth flyby in October 2013, two years after its launch on August 5, 2011.^[27] The spacecraft performed an orbit insertion burn to slow it enough to allow capture. It was expected to make three 53-day orbits before performing another burn on December 11, 2016, that would bring it into a 14-day polar orbit called the Science Orbit. Because of a suspected problem in *Juno*'s main engine, the burn scheduled on December 11, 2016, was cancelled and *Juno* remained in its 53-day orbit until the first Ganymede encounter of its Extended Mission.^[28] This extended mission began with a flyby of Ganymede on June 7, 2021.^{[29][30]} Subsequent flybys of Europa and then Io decreased the orbital period to 33 days by February 2024.^{[31][32]}

During the science mission, infrared and microwave instruments will measure the thermal radiation emanating from deep within Jupiter's atmosphere. These observations will complement previous studies of its composition by assessing the abundance and distribution of water, and therefore oxygen. This data will provide insight into Jupiter's origins. *Juno* will also investigate the convection that drives natural circulation patterns in Jupiter's atmosphere. Other instruments aboard *Juno* will gather data about its gravitational field and polar magnetosphere. The *Juno* mission was planned to conclude in February 2018 after completing 37 orbits of Jupiter, but now has been commissioned through 2025 to do a further 42 additional orbits of Jupiter as well as close flybys of Ganymede, Europa and Io.^[33] The probe was then intended to be deorbited and burnt up in Jupiter's outer atmosphere^{[5][6]} to avoid any possibility of impact and biological contamination of one of its moons.^[34]

Flight trajectory

Launch

Juno was launched atop an Atlas V (551 configuration) at Cape Canaveral Air Force Station (CCAFS), Florida on August 5, 2011, 16:25:00 UTC. The Atlas V (AV-029) used a Russian-built RD-180 main engine, powered by kerosene and liquid oxygen. The main engine ignited and underwent checkout then, 3.8 seconds later, the five strap-on solid rocket boosters (SRBs) ignited. Following the SRB burnout, about 93 seconds into the flight, two of the spent boosters fell away from the vehicle, followed 1.5 seconds later by the remaining three. When heating levels had dropped below predetermined limits, the payload fairing that protected *Juno* during launch and transit through the thickest part of the atmosphere separated, about 3 minutes 24 seconds into the flight. The Atlas V main engine cut off 4 minutes 26 seconds



Animation of Juno's trajectory from August 5, 2011

■ Juno · ■ Earth · ■ Mars · ■ Jupiter



Juno awaiting its launch in 2011

after liftoff. Sixteen seconds later, the Centaur second stage ignited, and it burned for about 6 minutes, putting the satellite into an initial parking orbit.^[35] The vehicle coasted for about 30 minutes, and then the Centaur was reignited for a second firing of 9 minutes, placing the spacecraft on an Earth escape trajectory in a heliocentric orbit.^[35]

Prior to separation, the Centaur stage used onboard reaction engines to spin *Juno* up to 1.4 r.p.m. About 54 minutes after launch, the spacecraft separated from the Centaur and began to extend its solar panels.^[35] Following the full deployment and locking of the solar panels, *Juno*'s batteries began to recharge. Deployment of the solar panels reduced *Juno*'s spin rate by two-thirds. The probe is spun to ensure stability during the voyage and so that all instruments on the probe are able to observe Jupiter.^{[34][15]}

The voyage to Jupiter took five years, and included two orbital maneuvers in August and September 2012 and a flyby of the Earth on October 9, 2013.^{[36][37]} When it reached the Jovian system, *Juno* had traveled approximately 19 astronomical units (2.8 billion kilometres).^[38]



Atlas V on launch pad

Lift-off

Launch video

Deep space maneuvers and flyby of the Earth

After traveling for about a year in an elliptical heliocentric orbit, *Juno* performed two deep space maneuvers (DSMs), firing its engine twice near aphelion (beyond the orbit of Mars) to change its orbit^[40] and return to pass by the Earth at a distance of 559 kilometers in October 2013.^[36] The combination of the DSMs and the resulting Earth flyby^[41] helped Juno slingshot itself toward the Jovian system in a maneuver called a gravity assist.^[42] The spacecraft received a boost in speed of more than 3.9 km/s (8,700 mph), and it was set on a course to Jupiter.^{[42][43][44]} The flyby was also used as a rehearsal for the *Juno* science team to test some instruments and practice certain procedures before the arrival at Jupiter.^[43]

Insertion into Jovian orbit

Jupiter's gravity accelerated the approaching spacecraft to around 210,000 km/h (130,000 mph).^[5] On July 5, 2016, between 03:18 and 03:53 UTC Earth-received time, an insertion burn lasting 2,102 seconds decelerated *Juno* by 542 m/s (1,780 ft/s)^[45] and changed its trajectory from a hyperbolic flyby to an elliptical, polar orbit with a period of about 53.5 days.^[46] The spacecraft successfully entered Jovian orbit on July 5, 2016, at 03:53 UTC.^[4]

Orbit and environment

Juno's highly elliptical initial polar orbit takes it within 4,200 km (2,600 mi) of the planet and out to 8.1 × 10⁶ km (5.0 × 10⁶ mi), far beyond Callisto's orbit. An eccentricity-reducing burn, called the Period Reduction Maneuver, was planned that would drop the probe into a much shorter 14 day science orbit.^[47]

Originally, *Juno* was expected to complete 37 orbits over 20 months before the end of its mission. Due to problems with helium valves that are important during main engine burns, mission managers announced on February 17, 2017, that *Juno* would remain in its original 53-day orbit, since the chance of an engine misfire putting the spacecraft into a bad orbit was too high.^[28] *Juno* completed only 12 science orbits before the end of its budgeted mission plan, ending July 2018.^[48] In June 2018, NASA extended the mission through July 2021.

The orbits were carefully planned in order to minimize contact with Jupiter's dense radiation belts, which can damage spacecraft electronics and solar panels, by exploiting a gap in the radiation envelope near the planet, passing through a region of minimal radiation.^{[9][49]} The Juno Radiation Vault, with 1-centimeter-thick titanium walls (three times as thick as the *Galileo* spacecraft body's), also aids in protecting *Juno*'s electronics by reducing the incoming radiation by a factor of 800.^[50] Despite the intense radiation, JunoCam and the Jovian Infrared Auroral Mapper (JIRAM) were designed to endure at least eight orbits, while the Microwave Radiometer (MWR) was made to endure at least eleven orbits. All instruments are operational as of perijove 71.^[51] Although the flux of electrons close to Jupiter is about ten times as high as it is around Jupiter's moon Europa,^[52] *Juno* will still receive a lower total dose of radiation in its polar orbit (20 Mrad through end of mission)^[53] than the *Galileo* orbiter received in its equatorial orbit. *Galileo*'s subsystems were damaged by radiation during its mission, including an LED in its data recording system.^[54]

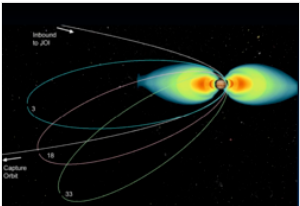
Orbital operations

The spacecraft completed its first flyby of Jupiter (perijove 1) on August 26, 2016, and captured the first images of the planet's north pole.^[55]

On October 14, 2016, days prior to perijove 2 and the planned Period Reduction Maneuver, telemetry showed that some of *Juno*'s helium valves were not opening properly.^[56] On October 18, 2016, some 13 hours before its second close approach to Jupiter, *Juno* entered into safe mode, an operational mode engaged when its onboard computer encounters unexpected conditions. The spacecraft powered down all non-critical systems and reoriented itself to face the Sun to gather the most power. Due to this, no science operations were conducted during perijove 2.^[57]

On December 11, 2016, the spacecraft completed perijove 3, with all but one instrument operating and returning data. One instrument, JIRAM, was off pending a flight software update.^[58] Perijove 4 occurred on February 2, 2017, with all instruments operating.^[28] Perijove 5 occurred on March 27, 2017.^[59] Perijove 6 took place on May 19, 2017.^{[59][60]}

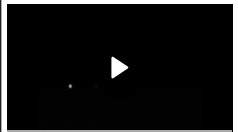
Although the mission's lifetime is inherently limited by radiation exposure, almost all of this dose was planned to be acquired during the perijoves. As of 2017, the 53.4 day orbit was planned to be maintained through July 2018 for a total of twelve science-gathering perijoves. At the end of this prime mission, the project was planned to go through a science review process by NASA's Planetary Science Division to determine if it will receive funding for an extended mission.^[28] In June 2018, NASA extended the mission operations plan to July 2021.^[61]



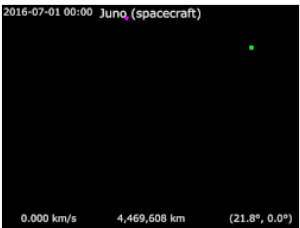
Juno's elliptical orbit and the Jovian radiation belts



South America ^[39] as seen by JunoCam on its October 2013 Earth flyby



Video of Earth and Moon taken by the Juno spacecraft



Animation of *Juno*'s trajectory around Jupiter from June 1, 2016, to October 1, 2028
■ Juno ■ Jupiter

Extended missions

In January 2021, NASA extended the mission operations to September 2025.^[62] In this phase *Juno* began to examine Jupiter's major moons, Ganymede, Europa and Io. A flyby of Ganymede occurred on June 7, 2021, 17:35 UTC, coming within 1,038 km (645 mi), the closest any spacecraft has come to the moon since *Galileo* in 2000.^{[29][30][63]} A flyby of Europa took place on September 29, 2022, at a distance of 352 km (219 mi).^{[64][65]} Juno performed two flybys of Io on December 30, 2023, and February 3, 2024, gathering observational data on volcanic activity. From April 2024, Juno will begin a series of experiments to learn more about Jupiter's interior shape and structure.^[66]

Juno's second extended mission (EM2) began in October 2025 with a planned three-year duration. Principal Investigator Scott Bolton noted "The new mission provides opportunities for Juno to unexplore new regions in the Jovian system, and to follow up on Juno's discoveries made during its prime and 1st extended missions. [...] During EM2, Juno will dive deep within Jupiter's inner radiation belts where the rings and inner moons reside. EM2 provides an opportunity for a thorough investigation of these components and their complex interaction, providing a unique data set to compare with other giant planet ring systems, including the ice giants."^[67]



Ganymede, photographed on 7 June 2021 by *Juno* during its extended mission

Planned deorbit and disintegration

NASA originally planned to deorbit the spacecraft into the atmosphere of Jupiter after completing 32 orbits of Jupiter. The controlled deorbit was intended to eliminate space debris and risks of contamination of possible life-bearing moons (especially Europa) by surviving terrestrial microorganisms onboard the spacecraft in accordance with NASA's planetary protection guidelines.^{[68][69][70]} NASA has since extended the mission twice, first to September 2025 and again to September 2028 and no deorbit is planned.^{[71][62][67]}

Team

Scott Bolton of the Southwest Research Institute in San Antonio, Texas is the principal investigator and is responsible for all aspects of the mission. The Jet Propulsion Laboratory in California manages the mission and the Lockheed Martin Corporation was responsible for the spacecraft development and construction. The mission is being carried out with the participation of several institutional partners. Coinvestigators include Toby Owen of the University of Hawaii, Andrew Ingersoll of California Institute of Technology, Frances Bagenal of the University of Colorado at Boulder, and Candy Hansen of the Planetary Science Institute. Jack Connerney of the Goddard Space Flight Center served as instrument lead.^{[72][73]}

Cost

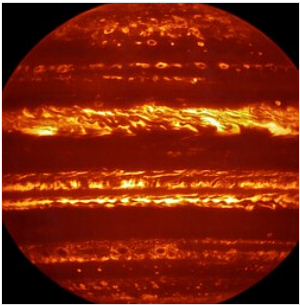
Juno was originally proposed at a cost of approximately US\$700 million (fiscal year 2003) for a launch in June 2009 (equivalent to US\$1225 million in 2025). NASA budgetary restrictions resulted in postponement until August 2011, and a launch on board an Atlas V rocket in the 551 configuration. As of 2019 the mission was projected to cost US\$1.46 billion for operations and data analysis through 2022.^[74]

Scientific objectives

The *Juno* spacecraft's suite of science instruments will:^[76]

- Determine the ratio of oxygen to hydrogen, effectively measuring the abundance of water in Jupiter, which will help distinguish among prevailing theories linking Jupiter's formation to the Solar System.

- Obtain a better estimate of Jupiter's core mass, which will also help distinguish among prevailing theories linking Jupiter's formation to the Solar System.
- Precisely map Jupiter's gravitational field to assess the distribution of mass in Jupiter's interior, including properties of its structure and dynamics.
- Precisely map Jupiter's magnetic field to assess the origin and structure of the field, and the depth at which the planet's magnetic field is created. This experiment will also help scientists understand the fundamental physics of dynamo theory.
- Map the variation in atmospheric composition, temperature, structure, cloud opacity and dynamics to pressures far greater than 100 bar (10 MPa; 1,500 psi) at all latitudes.
- Characterize and explore the three-dimensional structure of Jupiter's polar magnetosphere and auroras.^[76]
- Measure the orbital frame-dragging, known also as Lense–Thirring precession caused by the angular momentum of Jupiter,^{[77][78]} and possibly a new test of general relativity effects connected with the Jovian rotation.^[79]



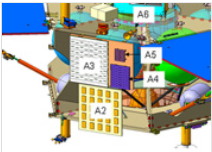
Jupiter imaged using the VISIR instrument on the Very Large Telescope, 2016. These observations helped to plan *Juno*'s mission.^[75]

Scientific instruments

The *Juno* mission's scientific objectives are being achieved with a payload of nine instruments on board the spacecraft.^{[80][81][82][83][84]}

Microwave radiometer (MWR)

The microwave radiometer comprises six antennas mounted on two of the sides of the body of the probe. They will perform measurements of electromagnetic waves on frequencies in the microwave range: 600 MHz, 1.2, 2.4, 4.8, 9.6 and 22 GHz, the only microwave frequencies which are able to pass through the thick Jovian atmosphere. The radiometer will measure the abundance of water and ammonia in the deep layers of the atmosphere up to 200 bar (20 MPa; 2,900 psi) pressure or 500–600 km (310–370 mi) deep. The combination of different wavelengths and the emission angle should make it possible to obtain a temperature profile at various levels of the atmosphere. The data collected will determine how deep the atmospheric circulation is.^{[85][86]} The MWR is designed to function through orbit 11 of Jupiter.^[87] (Principal investigator: Mike Janssen, Jet Propulsion Laboratory)



Microwave Radiometer

Jovian Infrared Auroral Mapper (JIRAM)

The spectrometer mapper JIRAM, operating in the near infrared (between 2 and 5 μm), conducts surveys in the upper layers of the atmosphere to a depth of between 50 and 70 km (31 and 43 mi) where the pressure reaches 5 to 7 bar (500 to 700 kPa). JIRAM will provide images of the aurora in the wavelength of 3.4 μm in regions with abundant H₃⁺ ions. By measuring the heat radiated by the atmosphere of Jupiter, JIRAM can determine how clouds with water are flowing beneath the surface. It can also detect methane, water vapor, ammonia and phosphine. It was not required that this device meets the radiation resistance requirements.^{[88][89][90]} The JIRAM instrument is expected to operate through the eighth orbit of Jupiter.^[87]

(Principal investigator: Alberto Adriani, Italian National Institute for Astrophysics)



Jovian Infrared Auroral Mapper

JIRAM's spin-compensation mirror is stuck since PJ44, but the instrument is operational.^[91]

Magnetometer (MAG)

The magnetic field investigation has three goals: mapping of the magnetic field, determining the dynamics of Jupiter's interior, and determination of the three-dimensional structure of the polar magnetosphere. The magnetometer experiment consists of the Flux Gate Magnetometer (FGM), which will observe the strength and direction of the magnetic field lines, and the Advanced Stellar Compass (ASC), which will monitor the orientation of the magnetometer sensors.^[82]
(Principal investigator: Jack Connerney, NASA's [Goddard Space Flight Center](#))



MAG

Gravity Science (GS)

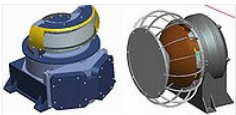
The purpose of measuring gravity by radio waves is to establish a map of the distribution of mass inside Jupiter. The uneven distribution of mass in Jupiter induces small variations in gravity all along the orbit followed by the probe when it runs closer to the surface of the planet. These gravity variations drive small probe velocity changes. The purpose of radio science is to detect the Doppler effect on radio broadcasts issued by *Juno* toward Earth in Ka-band and X-band, which are frequency ranges that can conduct the study with fewer disruptions related to the solar wind or Jupiter's ionosphere.^{[92][93][81]}
(Principal investigator: John Anderson, [Jet Propulsion Laboratory](#); Principal investigator (Juno's Ka-band Translator): Luciano Iess, [Sapienza University of Rome](#))



Gravity Science

Jovian Auroral Distributions Experiment (JADE)

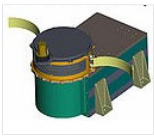
The energetic particle detector JADE will measure the angular distribution, energy, and the velocity vector of ions and electrons at *low* energy (ions between 13 eV and 20 KeV, electrons of 200 eV to 40 KeV) present in the aurora of Jupiter. On JADE, like JEDI, the electron analyzers are installed on three sides of the upper plate which allows a measure of frequency three times higher.^{[81][94]}
(Principal investigator: [David McComas](#), [Southwest Research Institute](#))



JADE

Jovian Energetic Particle Detector Instrument (JEDI)

The energetic particle detector JEDI will measure the angular distribution and the velocity vector of ions and electrons at *high* energy (ions between 20 keV and 1 MeV, electrons from 40 to 500 keV) present in the polar magnetosphere of Jupiter. JEDI has three identical sensors dedicated to the study of particular ions of hydrogen, helium, oxygen and sulfur.^{[81][95]}
(Principal investigator: [Barry Mauk](#), [Applied Physics Laboratory](#))



JEDI

Radio and Plasma Wave Sensor (Waves)

This instrument will identify the regions of auroral currents that define Jovian radio emissions and acceleration of the auroral particles by measuring the radio and plasma spectra in the auroral region. It will also observe the interactions between Jupiter's atmosphere and magnetosphere. The instrument consists of two antennae that detect radio and plasma waves.^[82]
(Principal investigator: [William Kurth](#), [University of Iowa](#))

Ultraviolet Spectrograph (UVS)

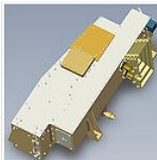
UVS will record the wavelength, position and arrival time of detected ultraviolet photons during the time when the spectrograph slit views Jupiter during each turn of the spacecraft. The instrument will provide spectral images of the UV auroral emissions in the polar magnetosphere.^[82]
(Principal investigator: [G. Randall Gladstone](#), [Southwest Research Institute](#))



Radio and Plasma Wave Sensor

JunoCam (JCM)

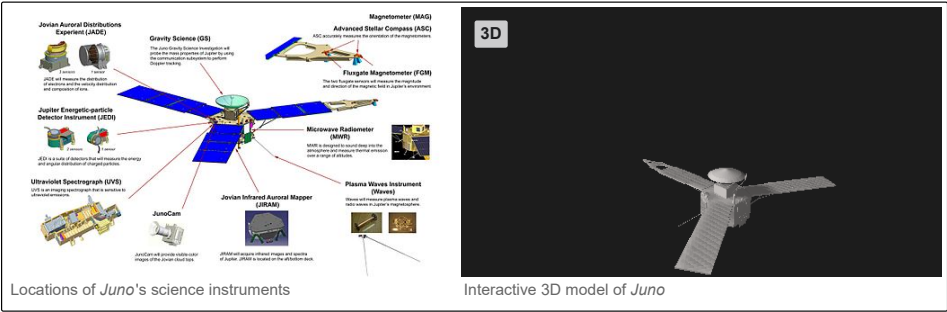
A visible light camera/telescope, included in the payload to facilitate education and public outreach; later re-purposed to study the dynamics of Jupiter's clouds, particularly those at the poles.^[96] It was anticipated that it would operate through only eight orbits of Jupiter ending in September 2017^[51] due to the planet's damaging radiation and magnetic field,^{[87][97]} during Juno's 47th orbit, the imager began showing hints of radiation damage. By orbit 56, nearly all the images were corrupted; the cause was identified as a damaged voltage regulator. By annealing the camera at a temperature of 25 °C (77 °F), the camera was brought back to operations. Junocam undergoes this operation periodically and as of July 2025 remains in operation.^[98]
(Principal investigator: [Michael C. Malin](#), [Malin Space Science Systems](#))



Ultraviolet Spectrograph



JunoCam



Operational components

Satellite bus

Juno's satellite bus, its main electronics and propulsion box, is a hexagonal prism.^[2]

Solar panels

Juno is the first mission to Jupiter and the Outer Solar System to use solar panels instead of the radioisotope thermoelectric generators (RTG) used by *Pioneer 10*, *Pioneer 11*, the Voyager program, *Ulysses*, *Cassini–Huygens*, *New Horizons*, and the *Galileo* orbiter.^{[99][100]} Once in orbit around Jupiter, *Juno* receives only 4% as much sunlight as it would on Earth, but the global shortage of plutonium-238 at the time,^{[101][102][103][104]} as well as advances made in solar cell technology over the past several decades, makes it economically preferable to use solar panels of practical size to provide power at a distance of 5 AU from the Sun.^[105]

The *Juno* spacecraft uses three solar panels symmetrically arranged around the spacecraft. Shortly after it cleared Earth's atmosphere, the panels were deployed. Two of the panels have four hinged segments each, and the third panel has three segments and a magnetometer. Each panel is 2.7 by 8.9 m (8 ft 10 in by 29 ft 2 in)^[106] providing 50 square metres (540 sq ft) of active cells^{[107][108]} – the largest on any NASA deep-space probe at the time of launching.^[17]

The combined mass of the three panels is nearly 340 kg (750 lb).^[15] If the panels were optimized to operate at Earth, they would produce 12 to 14 kilowatts of power. Only about 486 watts were generated when *Juno* arrived at Jupiter, projected to decline to near 420 watts as radiation degrades the cells.^[109] The solar panels will remain in sunlight continuously from launch through the end of the mission, except for short periods during the operation of the main engine and eclipses by Jupiter. A central power distribution and drive unit monitors the power that is generated by the solar panels and distributes it to instruments, heaters, and experiment sensors, as well as to batteries that are charged when excess power is available. Two 55 Ah lithium-ion batteries that are able to withstand the radiation environment of Jupiter provide power when *Juno* passes through eclipse.^[110]

Telecommunications

Juno uses in-band signaling ("tones") for several critical operations as well as status reporting during cruise mode,^[111] but it is expected to be used infrequently. Communications are via the 34 m (112 ft) and 70 m (230 ft) antennas of the NASA Deep Space Network (DSN) utilizing an X-band direct link.^[110] The command and data processing of the *Juno* spacecraft includes a flight computer capable of providing about 50 Mbit/s of instrument throughput. Gravity science subsystems use the X-band and K_a-band Doppler tracking and autoranging.^[112]

Due to telecommunications constraints, *Juno* will only be able to return about 40 megabytes of JunoCam data during each 11-day orbital period, limiting the number of images that are captured and transmitted during each orbit to somewhere between 10 and 100 depending on the compression level used.^[113] The overall amount of data downlinked on each orbit is significantly higher and used for the mission's scientific instruments; JunoCam is intended for public outreach and is thus secondary to the science data. This is comparable to the previous *Galileo* mission that orbited Jupiter, which captured thousands of images^[114] despite its slow data rate of 1000 bit/s (at maximum compression level) due to the failure of its high gain antenna.

The communication system is also used as part of the Gravity Science experiment.^[115]



Illumination test on one of *Juno*'s solar panels



Juno's high-gain antenna dish being installed

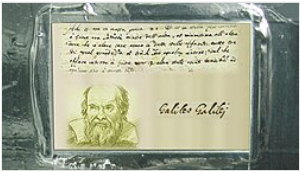
Propulsion

Juno uses a LEROS 1b main engine with hypergolic propellant, manufactured by Moog Inc in Westcott, Buckinghamshire, England.^[116] It uses approx. 2,000 kg (4,400 lb) of hydrazine and nitrogen tetroxide for propulsion, including 1,232 kg (2,716 lb) available for the Jupiter Orbit Insertion plus subsequent orbital maneuvers. The engine provides a thrust of 645 newtons. The engine bell is enclosed in a debris shield fixed to the spacecraft body, and is used for major burns. For control of the vehicle's orientation (attitude control) and to perform trajectory correction maneuvers, *Juno* utilizes a monopropellant reaction control system (RCS) consisting of twelve small thrusters that are mounted on four engine modules.^[110]

Galileo plaque and minifigures

Juno carries a plaque to Jupiter, dedicated to Galileo Galilei. The plaque was provided by the Italian Space Agency (ASI) and measures 7.1 by 5.1 cm (2.8 by 2.0 in). It is made of flight-grade aluminum and weighs 6 g (0.21 oz).^[117] The plaque depicts a portrait of Galileo and a text in Galileo's own handwriting, penned in January 1610, while observing what would later be known to be the Galilean moons.^[117] The text translates as:

On the 11th it was in this formation – and the star closest to Jupiter was half the size than the other and very close to the other so that during the previous nights all of the three observed stars looked of the same dimension and among them equally afar; so that it is evident that around Jupiter there are three moving stars invisible till this time to everyone.



Galileo Galilei plaque

The spacecraft also carries three Lego minifigures representing Galileo Galilei, the Roman god Jupiter, and his sister and wife, the goddess Juno. In Roman mythology, Jupiter drew a veil of clouds around himself to hide his mischief. Juno was able to peer through the clouds and reveal Jupiter's true nature. The Juno minifigure holds a magnifying glass as a sign of searching for the truth, and Jupiter holds a lightning bolt. The third Lego crew member, Galileo Galilei, has his telescope with him on the journey.^[118] The figurines were produced in partnership between NASA and Lego as part of an outreach program to inspire children's interest in science, technology, engineering, and mathematics (STEM).^[119] Although most Lego toys are made of plastic, Lego specially made these minifigures of aluminum to endure the extreme conditions of space flight.^[120]

Scientific results

Among early results, *Juno* gathered information about Jovian lightning that revised earlier theories.^[121] *Juno* provided the first views of Jupiter's north pole, as well as providing insight about Jupiter's aurorae, magnetic field, and atmosphere.^[122]

In 2021, analysis of the frequency of interplanetary dust impacts (primarily on the backs of the solar panels), as Juno passed between Earth and the asteroid belt, indicated that this dust, which causes the Zodiacal light, comes from Mars, rather than from comets or asteroids that come from the outer solar system, as was previously thought.^[123]

Juno made many discoveries that are challenging existing theories about Jupiter's formation. When Juno flew over the poles of Jupiter it imaged clusters of stable cyclones that exist at the poles.^[124] It found that the magnetosphere of Jupiter is uneven and chaotic. Using its Microwave Radiometer, Juno found that the red and white bands that can be seen on Jupiter extend hundreds of kilometers into the Jovian atmosphere, yet the interior of Jupiter is not evenly mixed. This has resulted in the theory that Jupiter does not have a solid core as previously thought, but a "fuzzy" core made of pieces of rock and metallic hydrogen. This peculiar core may be a result of a collision that happened early on in Jupiter's formation.^[125]

In April 2020, *Juno* detected a meteor impact on Jupiter, with estimated mass of 250–5000 kg.^[126]

Results from *Juno* on storms suggests that they are far taller than expected, with some extending 60 miles (100 kilometers) below the cloud tops and others, including the Great Red Spot, extending over 200 miles (350 kilometers). With *Juno* traveling low over Jupiter's cloud deck at about 130,000 mph (209,000 kph) Juno scientists were able to measure velocity changes as small as 0.01 millimeter per second using a NASA's Deep Space Network tracking antenna, from a distance of more than 400 million miles (650 million kilometers). This enabled the team to constrain the depth of the Great Red Spot to about 300 miles (500 kilometers) below the cloud tops. The new results show that the cyclones are warmer on top, with lower atmospheric densities, while they are colder at the bottom, with higher densities. Anticyclones, which rotate in the opposite direction, are colder at the top but warmer at the bottom.^[127]